Vitaran Edutech

HINTS & SOLUTIONS

INTEGRATED PROGRAM

Biology

Class – 8

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1 Cell- the basic unit of life

Solutions

Worksheet – 1

1. **Correct answer – C**
   Antonie Van Leeuwenhoek was the first to observe live cells like bacteria and protozoa using his handcrafted microscopes.
2. **Correct answer – B**
   Robert Hooke observed dead cell walls in a thin slice of cork and described them as "cells," meaning small rooms.
3. **Correct answer – C**
   This is not a postulate; for example, prokaryotic cells and mature human red blood cells lack a defined nucleus.
4. **Correct answer – B**
   This Latin phrase was popularized by Rudolf Virchow to explain the origin of new cells through division.
5. **Correct answer – A**
   Mycoplasmas are the smallest known living cells, typically measuring around 0.3 micrometers in length.
6. **Correct answer – D**
   The ostrich egg is a single cell and is currently recognized as the largest individual cell by volume.
7. **Correct answer – C**
   Bacteria are classified by shape; spirilla are spiral or helix-shaped, whereas cocci are spherical and bacilli are rod-shaped.
8. **Correct answer – B**
   An organ consists of multiple tissue types working together to perform a complex function within an organism
9. **Correct answer – C**
   The provided text states that cell walls, chloroplasts, and large vacuoles are characteristic of plant cells, leaving centrioles as the animal-specific structure.
10. **Correct answer – B**
    Unlike unicellular organisms that perform all functions in one cell, multicellular organisms assign specific tasks to different cell types.

worksheet – 2

1. **Correct answer – B**
   Mesosomes are internal extensions of the plasma membrane in prokaryotes that increase surface area for enzymatic activities like respiration and secretion.
2. **Correct answer – C**
   Inclusion bodies (or ergastic substances) are non-living structures in the cytoplasm, such as phosphate granules or glycogen granules, used to store reserve materials.
3. **Correct answer – A**
   Jonathan Singer and Garth Nicolson proposed this model to describe the cell membrane as a fluid lipid bilayer with a "mosaic" of various proteins.
4. **Correct answer – B**
   This arrangement protects the hydrophobic (water-fearing) tails from the aqueous environment inside and outside the cell.
5. **Correct answer – D**
   Integral proteins are embedded within the lipid bilayer, whereas peripheral proteins are loosely attached to the membrane surface.
6. **Correct answer – C**
   The middle lamella is composed primarily of calcium and magnesium pectates, which glue adjacent plant cells together.

7. **Correct answer – B**

   These are microscopic channels that traverse the cell walls of plant cells, enabling transport and communication between them.

8. **Correct answer – C**

   Unlike plant cell walls made of cellulose, fungal cell walls are made of chitin, a tough nitrogen-containing polysaccharide.

9. **Correct answer – A**

   The primary wall is the first layer formed; it is thin, flexible, and capable of expansion as the plant cell grows.

10. **Correct answer – C**

    Once a cell stops growing, it may deposit a secondary wall inside the primary wall, providing significant structural strength and rigidity.

worksheet 3

1. **Correct answer – B**

   Protoplasm is defined as the entire living substance of the cell, comprising the nucleus and the surrounding cytoplasm.

2. **Correct answer – A**

   Robert Brown observed and named the nucleus while studying orchid cells in 1831.

3. **Correct answer – C**

   The nucleolus is the specific dense region within the nucleus where ribosomal RNA is synthesized and ribosomes are assembled.

4. **Correct answer – C**

   The "rough" appearance is due to numerous ribosomes attached to the outer surface of the membrane for protein synthesis.

5. **Correct answer – B**

   Smooth Endoplasmic Reticulum (SER) lacks ribosomes and specializes in synthesizing lipids and metabolizing carbohydrates/glycogen.

6. **Correct answer – C**
   Mitochondria convert chemical energy from food into ATP, the primary energy currency used by cells.

7. **Correct answer – B**
   Prokaryotic ribosomes are small, non-membrane bound organelles measuring approximately 15 to 20 nanometers.

8. **Correct answer – B**
   It modifies, sorts, and packages proteins and lipids into vesicles for delivery to specific destinations.

9. **Correct answer – C**
   If a cell is damaged, lysosomes may burst, and their digestive enzymes will consume the cell itself.

10. **Correct answer – C**
    Aleuroplast: These specialized leucoplasts are responsible for the storage of proteins, typically found in seeds.

Competitive Corner

SINGLE OPTION CORRECT TYPE

1. **Correct answer – A**
   The Golgi complex receives products from the ER to modify and package them into secretory vesicles.

2. **Correct answer – C**
   Amoeba uses these "false feet" (cytoplasmic extensions) for locomotion and capturing food.

3. **Correct answer – C**
   Animal cells lack cell walls and chloroplasts, which are present in the other options.

4. **Correct answer – A**
   This correctly labels the flow of materials from the *cis* face to the *trans* face of the Golgi.

5. **Correct answer – D**

   A-Incoming transport vesicle, B-Lumen, C-Secretory vesicle, D-Cisternae, E-Newly forming vesicle Chromosomes carry genes and are visible in the nucleus during division, but prokaryotes have a "nucleoid" rather than true organized chromosomes.

6. **Correct answer – C**

   Zacharias Janssen is credited with inventing the first compound microscope around 1595.

7. **Correct answer – A**

   The image shows a mitochondrion, which contains the enzymes necessary for aerobic respiration and ATP production.

8. **Correct answer – D**

   Sclerenchyma tissues consist of dead cells with thick, lignified walls that provide mechanical support to plants.

9. **Correct answer – A**

   This follows the logical order: extract the sample, place it on the slide, stain it, and finally observe it.

10. **Correct answer – D**

    This correctly identifies the internal fluid (stroma), the double membrane system, and the connecting lamella.

ONE OR MORE THAN ONE OPTION CORRECT TYPE

11. **Correct answer – C, D**

    Mature human red blood cells lose their nucleus to carry more oxygen and never contain plastids (found only in plants).

12. **Correct answer – A, B**

    Protoplasm includes everything inside the cell membrane, and chloroplasts are the green plastids used for photosynthesis.

13. **Correct answer – C, D**

    Plant cells possess a rigid cell wall and a large central vacuole, both of which are generally absent or much smaller in animal cells.

14. **Correct answer – A**

    Purkinje named protoplasm, Schwann co-proposed cell theory, and Virchow stated cells come from pre-existing cells; Hooke discovered the cell, not Leeuwenhoek.

15. **Correct answer – C, D**

    The cell wall provides structural rigidity and shape, while mitochondria generate energy through cellular respiration.

ASSERTION AND REASNING TYPE

16. **Correct answer – A**

    Chloroplasts are the "kitchen" because the chlorophyll within them captures light to synthesize food.

17. **Correct answer – C**

    Leeuwenhoek was the first to see "animalcules" (living cells), and Brown discovered the nucleus in 1831.

18. **Correct answer – A**

    A high surface-area-to-volume ratio in small cells allows for efficient nutrient exchange and internal signaling.

19. **Correct answer – C**

    While SER lacks ribosomes, it is involved in lipid synthesis, whereas RER is involved in protein synthesis.

MATCH THE FOLLOWING TYPE

20. **Correct answer – A**

    a-P, b-S, c-Q, d-R

21. **Correct answer – C**

    a-Q, b-S, c-P, d-R

COMPREHENSHION TYPE

Passage1

22. **Correct answer – C**

    The DNA inside the nucleus acts as a blueprint for all cellular functions and protein production.

23. **Correct answer – B**

    Deoxyribonucleic acid is the molecule that carries the genetic instructions in eukaryotic organisms.

24. **Correct answer – B**

    The nucleolus is the specialized site where the components of ribosomes are produced.

Passage2

25. Correct answer – B

The Golgi apparatus acts as a shipping center, preparing molecules for their final destinations.

26. Correct answer – B

Lysosomes contain digestive enzymes that break down debris or, if the cell is dying, the cell itself.

27. Correct answer – C

Chloroplasts contain chlorophyll to convert solar energy into chemical energy (glucose) for the plant.

Previous Year Questions

1. Correct answer – C

The organelle is the Golgi apparatus, which is involved in packaging and secretion; protein and lipid synthesis is the function of the Endoplasmic Reticulum.

2. Correct answer – D

Q represents mitochondria, which are present in both plant and animal cells and are responsible for producing ATP through cellular respiration.

3. Correct answer – B

Y represents a plant cell, and the statement is incorrect because plastids (like chloroplasts) are actually present in plant cells to perform photosynthesis.

4. Correct answer – B

Mitochondria are semiautonomous and absent in RBCs; Ribosomes are found in cytoplasm/ER; Golgi forms lysosomes; and Lysosomes digest microorganisms.

5. Correct answer – D

RBC, WBC, and Sperm are cells; Blood, Muscle, and Skin are tissues; Brain, Heart, Lung, and Kidney are organs.

6. Correct answer – D

In the paragraph, Large (plant vacuole) and Small (animal vacuole) are correct, but mitochondria should be replaced with ribosomes as mitochondria are present in both, not just animals.

7. Correct answer – D

Statement D is incorrect because P (Mitochondria) consumes oxygen for energy, while Q (Chloroplast) liberates oxygen during photosynthesis.

8. Correct answer – C

Q refers to a unicellular eukaryotic parasite; Plasmodium (malaria) and Entamoeba (dysentery) fit this description perfectly.

9. Correct answer – D

In this diagram, S is the Golgi apparatus (secretory) and R is a plastid/chloroplast (absent in animals); however, usually, in these specific Olympiad diagrams, ii and iv are tested as incorrect if labels are swapped.

2 MICROORGANISMS

Solutions

Worksheet – 1

1. **Correct answer – B**
   Robert Hooke – In 1665, Hooke published *Micrographia*, which featured the first detailed sketches of a microorganism, specifically the stalk of the fungus *Mucor*.
2. **Correct answer – C**
   1878 – The term 'microbe' was proposed by Charles Emmanuel Sédillot during a presentation at the French Academy of Sciences in 1878.
3. **Correct answer – D**
   300 times – Leeuwenhoek's superior hand-ground lenses allowed for magnifications up to 300x, enabling him to see "animalcules" for the first time.
4. **Correct answer – A**
   Thiomargarita namibiensis – This "Sulfur Pearl of Namibia" is a giant among bacteria, reaching 0.75 mm, which is large enough to be seen without a microscope.
5. **Correct answer – D**
   Bacteria are prokaryotic organisms, meaning they lack a membrane-bound nucleus (pseudo-nucleus) and other complex organelles.
6. **Correct answer – C**
   Unlike plants, fungi cannot make their own food and must obtain nutrients from organic matter or other organisms.
7. **Correct answer – B**
   Algae are simple autotrophic organisms with a "thallus" body, meaning they do not have specialized roots, stems, or leaves.
8. **Correct answer – A**
   Beneficial bacteria in the human gut, known as microflora, are essential for synthesizing Biotin and Vitamin K.
9. **Correct answer – C**
   They increase fertility by decomposing biomass through digestion – Soil microfauna break-down organic materials, which recycles nutrients and improves soil health.
10. **Correct answer – D**
    While extremely small, krill are classified as microfauna because they are minute-sized invertebrates.

worksheet – 2

1. **Correct answer – C**
   Rhinovirus Rhino (meaning nose) viruses are the most common viral infectious agents in humans and the primary cause of the common cold.
2. **Correct answer – A**
   Whooping cough *Bordetella pertussis* is the bacterium that colonizes the respiratory tract and causes the severe coughing fits known as pertussis or whooping cough.
3. **Correct answer – A**
   Acetobacter malorum Specific species of *Acetobacter* are used in the processing of coffee cherries to help break down the pulp through fermentation.
4. **Correct answer – C**
   Cow The word comes from the Latin 'vacca' because the first vaccine was derived from the cowpox virus.
5. **Correct answer – B**
   Cowpox Jenner noticed that milkmaids who had contracted cowpox were immune to the much more deadly smallpox virus.

6. **Correct answer – C**

Viral diseases Antibiotics are designed to target bacterial structures (like cell walls); therefore, they are generally ineffective against viruses.

7. **Correct answer – B**

Bioaugmentation is the practice of adding specific, lab-grown microorganisms to an environment to accelerate the detoxification of pollutants.

8. **Correct answer – C**

Paracoccus denitrificans This bacterium is highly effective at removing nitrates and organic pollutants from industrial and pharmaceutical wastewater.

9. **Correct answer – C**

Root nodules Leguminous plants (like peas and beans) form a symbiotic relationship with nitrogen-fixing bacteria within specialized swellings on their roots called nodules.

10. **Correct answer – B**

Pathogens The term "pathogen" comes from the Greek words meaning "producer of suffering" and specifically refers to any disease-causing microbe.

worksheet 3

1. **Correct answer – B**

It is an acute disease because it lasts for a very short duration.

2. **Correct answer – C**

This specific bacterium is the causative agent of typhoid fever in humans.

3. **Correct answer – B**

Ringworm is a fungal infection characterized by itchy, circular skin lesions.

4. **Correct answer – D**

Malaria is caused by the *Plasmodium* parasite, which belongs to the protozoa group.

5. **Correct answer – A**

This bacterial disease causes yellow, lesion-like spots on citrus leaves and fruits.

6. **Correct answer – B**

Puccinia graminis is the fungal pathogen responsible for reddish-brown spots on wheat.

7. **Correct answer – C**

TMV interferes with chlorophyll, creating a distinctive patchy or mottled appearance.

8. **Correct answer – C**

He developed the pasteurization process to kill microbes by heating and rapid cooling.

9. **Correct answer – C**

MAP replaces oxygen with gases like nitrogen to inhibit aerobic spoilage.

10. **Correct answer – B**

Enzymes are natural biological catalysts that trigger the chemical breakdown and rotting of food.

Competitive Corner

SINGLE OPTION CORRECT TYPE

1. **Correct answer – A**

Lactobacillus is the primary bacterium responsible for converting milk into curd.

2. **Correct answer – C**

Yeast is added to dough because it produces carbon dioxide, making the bread rise.

3. **Correct answer – A**

Vibrio cholerae is an example of type S. "S" typically represents comma-shaped (vibrio) bacteria which cause cholera.

4. **Correct answer – D**

It is a parasitic fungus. Bread mould (*Rhizopus*) is a saprophyte that feeds on dead matter, not a parasite.

5. **Correct answer – A**

   W – Virus, X – Protozoan, Y – Fungus – W is a bacteriophage, X is an amoeba, and Y is a bread mould.

6. **Correct answer – A**

   X – Bacteria, Y – Fungi, Z – Virus – Citrus canker is bacterial, wheat rust is fungal, and mosaic disease is viral.

7. **Correct answer – B**

   X – *Saccharomyces*; Y – Fermentation – *Saccharomyces* (yeast) converts sugar to alcohol through the fermentation process.

8. **Correct answer – A**

   Viruses are unique because they are inert outside a host and only become active inside living cells.

9. **Correct answer – C**

   Bacteria are prokaryotes and lack a membrane-bound nucleus, which the student observed.

10. **Correct answer – A**

    P-Air, Q-Tuberculosis, R-Bacteria, S-Malaria – This correctly matches the transmission, disease names, and organism types.

ONE OR MORE THAN ONE OPTION CORRECT TYPE

11. **Correct answer – C, D**

    Lactobacillus (not cyanobacteria) forms curd, and *Spirogyra* is an alga, not a fungus.

12. **Correct answer – A, B, C**

    Fungi, bacteria, and protozoa are matched correctly to their cellular structures; viruses are not multicellular.

13. **Correct answer – C, D**

    Malaria and Amoebiasis are the only two diseases in this list caused by protozoan parasites.

14. **Correct answer – A, B, C**

    Microorganisms have helpful and harmful roles and are generally too small for the naked eye.

15. **Correct answer – D**

    1, 2 & 3 – Proper storage, refrigeration, and careful handling are all critical to preventing food poisoning.

ASSERTION AND REASNING TYPE

16. **Correct answer – A**

    Both A and R are true. The CO2 released by yeast fermentation is exactly what causes the dough to expand.

17. **Correct answer – C**

    Both statements accurately define the biological nature and the process of immunization.

18. **Correct answer – A**

    Both A and R are true... – Lichens are the classic example of a symbiotic relationship between algae and fungi.

19. **Correct answer – A**

    Both A and R are true... – The lack of independent reproduction is the primary reason viruses are often classed as non-living.

MATCH THE FOLLOWING TYPE

20. **Correct answer – B**

    a-T, b-R, c-P, d-Q, e-S

21. **Correct answer – A**

    A) a-T, b-P, c-R, d-Q, e-S

COMPREHENSHION TYPE

Passage1

22. **Correct answer – B**

    Single-celled prokaryotes with pseudo-nuclei – As stated in the passage, bacteria lack a true, organized nucleus.

23. **Correct answer – A**

    Dark, moist places – The passage mentions that fungi thrive specifically in these types of environments.

24. Correct answer – D

Aquatic photoautotrophs – Algae are defined in the text as water-dwelling organisms that produce their own food.

Passage2

25. Correct answer – B

Microorganisms, enzymes, and oxidation – These are identified in the passage as the three main causes of spoilage.

26. Correct answer – C

Bacteria, yeasts, and molds – The text specifically lists these three groups as the microbial causes of food going bad.

27. Correct answer – C

Rotting – The passage explains that enzymes are the biological molecules responsible for the rotting process.

Previous Year Questions

1. Correct answer – D

Organism X (a Virus) is considered a connecting link because it shows characteristics of living things only inside a host cell and behaves like a non-living crystal outside.

2. Correct answer – D

Lactobacillus and Streptococcus are the primary bacteria used in the fermentation and curdling process required to produce cheese.

3. Correct answer – D

Cholera, Anthrax, Tuberculosis, and Typhoid are all caused by different types of bacteria, whereas the other options contain viral or protozoan diseases.

4. Correct answer – C

The statement is incorrect because Citrus canker is a bacterial disease caused by Xanthomonas citri, not a viral one.

5. Correct answer – C

P is a Protozoa (Entamoeba histolytica) that causes amoebic dysentery, and R is Tetanus, which is caused by bacteria entering through contaminated wounds.

6. Correct answer – D

In Paramecium (Q), cilia are used for both movement and capturing food, while in Amoeba (P), pseudopodia serve both these functions.

7. Correct answer – C

Statements P, Q, and R are false (fungi can reproduce sexually, algae can be unicellular, and Vibrio causes cholera), while S is true as many antibiotics are derived from these microbes.

8. Correct answer – C

This option correctly matches Typhoid to bacteria/water, Measles to virus/air, Polio to virus/water, and Dysentery to protozoa/water.

9. Correct answer – D

W is Air (TB transmission), X is Virus (Polio agent), Y is Cholera (bacterial/water disease), and Z is Protozoa (Malaria agent).

10. Correct answer – C

Matches (i), (ii), and (iv) are incorrect because Streptomyces provides antibiotics, Lactobacillus (not Rhizobium) curdles milk, and Euglena is a protist, not a nitrogen-fixing bacterium.

3 Reproduction in Animals

Solutions

Worksheet – 1

1. **Correct answer – B**
   Reproduction allows a species to persist over time by replacing individuals that die or age out of the population.
2. **Correct answer – C**
   Asexual reproduction involves a single parent, resulting in offspring that are genetically identical to that parent.
3. **Correct answer – B**
   Every form of reproduction requires the copying of genetic material so that the offspring receive the necessary biological instructions.
4. **Correct answer – C**
   Oviparous animals, such as birds and most reptiles, protect their developing embryos within eggs laid outside the body.
5. **Correct answer – B**
   Viviparous animals give birth to live young that receive direct nourishment from the mother's body during development.
6. **Correct answer – C**
   After the nucleus divides, the cell undergoes cytokinesis, where the membrane pinches inward to separate the two new cells.
7. **Correct answer – C**
   Plasmodium is a classic example of multiple fission, where the parent cell divides into many daughter cells at the same time.
8. **Correct answer – B**
   In organisms like Hydra, the bud is a physical outgrowth that relies on the parent's systems until it is mature enough to detach.
9. **Correct answer – B**
   Exogenous budding (like in Hydra) happens on the outside of the organism, while endogenous budding (like in sponges) happens internally.
10. **Correct answer – B**
    Fragmentation occurs when an organism breaks into distinct pieces, each of which has the regenerative ability to grow into a full adult.

worksheet – 2

1. **Correct answer – B**
   Hermaphrodites, such as earthworms, possess both male and female reproductive organs and can produce both types of gametes.
2. **Correct answer – C**
   The production of healthy sperm requires a temperature that is approximately 2-3 C lower than the internal body temperature.
3. **Correct answer – D**
   The acrosome is a cap-like structure at the tip of the sperm head that releases enzymes to digest the outer protective layers of the egg.
4. **Correct answer – C**
   Fertilization typically happens in the fallopian tube, where the sperm meets the egg after it has been released from the ovary.
5. **Correct answer – B**
   The uterus is the muscular organ where a fertilized egg (zygote) attaches itself and develops into a fetus until birth.

6. **Correct answer – C**

   Ovulation is the specific phase of the menstrual cycle when a mature egg is released from the ovary into the fallopian tube.

7. **Correct answer – B**

   The middle piece acts as the "powerhouse" of the sperm, containing numerous mitochondria that generate the energy needed for swimming.

8. **Correct answer – B**

   Fluid from the seminal vesicles and prostate gland provides nutrients like fructose and an alkaline environment to keep sperm active and safe.

9. **Correct answer – C**

   Although millions of sperm are released, a chemical change in the egg's membrane occurs immediately after the first sperm enters, blocking all others.

10. **Correct answer – C**

    Testosterone is the primary male sex hormone produced by the testes, responsible for sperm development and physical male traits.

worksheet 3

1. **Correct answer – B**

   External fertilisation occurs when gametes fuse in an aquatic medium outside the mother's body.

2. **Correct answer – B**

   Fertilisation forms a zygote, which divides to become an embryo before embedding in the uterine lining.

3. **Correct answer – C**

   The endometrium thickens to provide a nutrient-rich "bed" for the potential implantation of a zygote.

4. **Correct answer – C**

   The foetus is the developmental stage where all major body parts, such as limbs and a head, become clearly identifiable.

5. **Correct answer – C**

   The placenta acts as a life-support system for exchanging oxygen and nutrients and removing waste between the mother and foetus.

6. **Correct answer – B**

   Metamorphosis involves rapid transformations from a larval form (like a tadpole) to an adult form (like a frog).

7. **Correct answer – B**

   Tadpoles are adapted for aquatic life with gills and a tail, whereas adult frogs develop lungs and legs.

8. **Correct answer – B**

   This adaptation allows the adult frog to see above the water line while most of its body remains submerged.

9. **Correct answer – B**

   The silk moth transforms through distinct larva and pupa stages that differ significantly from the adult form.

10. **Correct answer – B**

    This is the average duration for complete human fetal development within the uterus.

Competitive Corner

SINGLE OPTION CORRECT TYPE

1. **Correct answer – A**

   The text notes reproduction ranges from asexual to sexual modes.

2. **Correct answer – B**

   Asexual reproduction produces clones that are genetically identical to the parent.

3. **Correct answer – C**

   Fragmentation allows an organism to grow into a new individual from a piece of its parent body.

4. **Correct answer – B**

   Oviparous embryos (chicks) use yolk for food, while viviparous embryos (humans) use a placenta.

5. **Correct answer – C**

   Internal buds that stay protected inside the parent are characteristic of endogenous budding.

6. **Correct answer – B**

   Mitochondria in the middle piece provide the energy (ATP) required for the sperm's tail to move.

7. **Correct answer – C**

   Blocking the fallopian tubes prevents fertilisation, as this is where the sperm and egg typically meet.

8. **Correct answer – B**

   Fructose in semen serves as nourishment to facilitate sperm movement.

9. **Correct answer – B**

   The thickening of the endometrium prepares the uterus to support an embryo.

10. **Correct answer – C**

    This is the defining characteristic that distinguishes a foetus from an earlier embryo.

ONE OR MORE THAN ONE OPTION CORRECT TYPE

11. **Correct answer – A, B, D**

    Exogenous buds are on the surface, endogenous are internal (like in sponges), and they typically derive nutrients from the parent initially.

12. **Correct answer – B, C, D**

    Viviparous embryos develop internally via a placenta, while oviparous embryos rely on yolk in external eggs.

13. **Correct answer – A, C, D**

    The male system includes testes, scrotum, sperm ducts, penis, and accessory glands; oviducts are part of the female system.

14. **Correct answer – A, B, C, D**

    The ovum is the largest human cell, spherical, a single cell, and released via ovulation.

15. **Correct answer – A, C**

    Humans and hens undergo internal fertilisation (hens then lay eggs), while frogs and fish typically use external fertilisation.

ASSERTION AND REASNING TYPE

16. **Correct answer – C**

    Amoeba uses binary fission, but binary fission results in *two* daughter cells, not "many" (that would be multiple fission).

17. **Correct answer – D**

    The egg is released from the *ovary*, not the uterus; the uterus is where implantation occurs.

18. **Correct answer – A**

    The acrosome uses enzymes like acrosin to penetrate the egg.

19. **Correct answer – D**

    Tadpoles cannot survive on land because they have gills and tails adapted for water.

MATCH THE FOLLOWING TYPE

20. **Correct answer – A**

    a-Q, b-R, c-S, d-P, e-T

21. **Correct answer – C**

    a-R, b-S, c-P, d-T, e-Q

COMPREHENSHION TYPE

Passage1

22. **Correct answer – B**
23. **Correct answer – C**
24. **Correct answer – B**

Passage2

25. **Correct answer – C**
26. **Correct answer – B**
27. **Correct answer – C**

Previous Year Questions

1. **Correct answer – B**

   In the Venn diagram (based on the provided image), **Y** refers to the sperm duct, but the statement describes it incorrectly relative to the urethra's position or transport.

2. **Correct answer – D**

   The scrotum (**T**) protects the testes (**P**) and maintains a lower temperature outside the abdomen.

3. **Correct answer – D**

   The resulting structure after fertilisation is a **zygote**, not a foetus.

4. **Correct answer – C**

   W (Viviparous) could be a bat; Y (Oviparous/External) could be a salmon.

5. **Correct answer – B**

   P (ovary) is in the abdomen; S (vagina) is the birth canal; R (uterus) is the womb; Q is the fallopian tube (fertilisation site, not implantation).

6. **Correct answer – D**

   Animal X could be crocodile whereas animal Y could be dog. Crocodiles lay eggs (oviparous), and dogs give live birth (viviparous).

7. **Correct answer – B**

   Plasmodium, Hermaphrodite, Human beings.

8. **Correct answer – B**

   (i), (iii), (v) and (viii) only Silk moths, frogs, butterflies, and cockroaches undergo metamorphosis.

9. **Correct answer – B**

   Amoeba, Hydra, Planaria.

10. **Correct answer – D**

    Q is the fallopian tube/uterus area; the vagina receives the penis.

4 Reaching the Age of Adolescence

Solutions

Worksheet – 1

1. **Correct answer – B**
   Adolescence marks the transitional stage of physical and psychological development from childhood to adulthood.
2. **Correct answer – C**
   This period encompasses the onset of puberty and continues until an individual reach full physical maturity, often called the "teenage" years.
3. **Correct answer – B**
   Endocrine glands do not use tubes (ducts); they rely on the circulatory system to transport hormones to target organs.
4. **Correct answer – C**
   Exocrine glands, such as sweat and salivary glands, use a system of ducts to release their secretions onto a surface or into a cavity.
5. **Correct answer – B**
   Hormones are signaling molecules that travel through the blood to tell specific parts of the body how to function or grow.
6. **Correct answer – C**
   Secreted by the pituitary gland, this hormone is essential for the healthy development of height, bone density, and muscle mass.
7. **Correct answer – B**
   A lack of thyroxine (often caused by iodine deficiency) leads to the abnormal enlargement of the thyroid gland in the neck.
8. **Correct answer – D**
   These glands secrete adrenaline, which increases heart rate and blood flow to prepare the body for sudden physical action.
9. **Correct answer – B**
   The presence of thyroxine in water is critical for tadpoles to undergo the biological transformation into adult frogs.
10. **Correct answer – C**
    Progesterone is specifically responsible for preparing and maintaining the lining of the uterus to support a potential pregnancy.

worksheet – 2

1. **Correct answer – B**
   This stage is defined by the transition from a child to an adult, involving significant physical and mental changes.
2. **Correct answer – C**
   Secreted by the testes, this hormone triggers changes like facial hair growth and deepening of the voice in boys.
3. **Correct answer – D**
   During puberty, the larynx (voice box) grows larger and protrudes from the neck, particularly in males.
4. **Correct answer – B**
   Higher hormone levels cause these glands to produce more oil (sebum), which can clog pores and lead to acne.
5. **Correct answer – C**
   This is the specific phase of the menstrual cycle where an ovary releases an egg for potential fertilization.
6. **Correct answer – B**
   Marking the beginning of reproductive maturity, this is the very first menstrual period a girl experiences.

7. **Correct answer – C**

   The hips and pelvic area widen in girls during puberty to prepare the body for future childbearing.

8. **Correct answer – B**

   When fertilization doesn't occur, the thickened lining of the uterus is no longer needed and is shed from the body.

9. **Correct answer – C**

   This stage, known as menopause, marks the end of a woman's reproductive phase and the stopping of the menstrual cycle.

10. **Correct answer – C**

    Despite various cultural myths, menstruation is a healthy, natural sign that the female reproductive system is functioning correctly.

worksheet 3

1. **Correct answer – C**
   In India, the legal marriageable age is 21 for boys and 18 for girls to ensure physical and mental maturity.
2. **Correct answer – D**
   Early pregnancy poses biological risks to the mother, increases stress, and often curtails educational opportunities.
3. **Correct answer – B**

   Human cells contain a total of 46 chromosomes, which are organized into 23 distinct pairs.

4. **Correct answer – C**

   Since mothers only have X chromosomes, the gender is determined by whether the father's sperm carries an X or a Y.

5. **Correct answer – B**

   Male semen naturally contains an equal distribution of X-type and Y-type sperm cells.

6. **Correct answer – C**

   Zinc is a vital micronutrient specifically linked to the proper development of reproductive organs and growth during puberty.

7. **Correct answer – B**

   Proper disposal in a bin prevents the spread of infections and keeps the environment clean.

8. **Correct answer – C**

   Proper disposal in a bin prevents the spread of infections and keeps the environment clean.

9. **Correct answer – B**

   HIV is transmitted through direct contact with infected blood, often via shared needles.

10. **Correct answer – C**

    These are protective nutrients that regulate body processes and boost immunity without being primary energy sources.

Competitive Corner

SINGLE OPTION CORRECT TYPE

1. **Correct answer – C**

   The hypothalamus monitors body conditions and signals the pituitary gland to release or stop hormones.

2. **Correct answer – C**

   The pancreas produces insulin; a lack of it leads to high blood glucose levels (diabetes).

3. **Correct answer – C**

   Iodine is essential for the thyroid to produce thyroxine, which regulates the body's metabolic rate.

4. **Correct answer – C**

   This is the fundamental structural difference between the two gland types.

5. **Correct answer – B**

   The "cracking" is a temporary lack of control over the vocal cords as the voice box enlarges.

6. **Correct answer – B**

   Different body parts grow at different rates, leading to temporary physical disproportion.

7. **Correct answer – A**

   It is a biological cleansing process, not a sign of being "unclean" or "sick."

8. **Correct answer – B**

   Physical activity releases endorphins and provides a healthy outlet for stress and insecurity.

9. **Correct answer – B**

   Scientifically, the father's sperm is the deciding factor for the child's sex.

10. **Correct answer – B**

    Pregnancy during growth years diverts essential nutrients away from the mother's own developing body.

ONE OR MORE THAN ONE OPTION CORRECT TYPE

11. **Correct answer – B, C, D**

    Hormones are required in *small* amounts, act on distant target organs, and come from ductless glands.

12. **Correct answer – B, C, D**

    While girls often start their spurt earlier, both sexes grow through bone lengthening until roughly age 18.

13. **Correct answer – A, C, D**

    Menstruation involves shedding the lining and the egg over several days; pregnancy only occurs if fertilization does happen.

14. **Correct answer – A, C**

    Sweat and salivary glands use ducts, whereas the thyroid and pituitary are ductless endocrine glands.

15. **Correct answer – A, B, C**

    Child marriage is a "social evil" because it negatively impacts health, mental well-being, and future life prospects.

ASSERTION AND REASNING TYPE

16. **Correct answer – A**

    The pituitary is the "master" because it directs the activity of other glands.

17. **Correct answer – D**

    Menstruation is a natural biological process, and the idea of "impurity" is a myth.

18. **Correct answer – C**

    Iodine is needed for the thyroid gland (thyroxine), not the adrenal glands.

19. **Correct answer – C**

    Body parts grow at different rates, which is exactly why adolescents can look awkward.

MATCH THE FOLLOWING TYPE

20. **Correct answer – A**

    a-R, b-S, c-P, d-Q

21. **Correct answer – D**

    a-Q, b-P, c-S, d-R, e-T, f-U

COMPREHENSHION TYPE

Passage1

22. **Correct answer – B**
23. **Correct answer – D**
24. **Correct answer – C**

Passage2

25. **Correct answer – C**
26. **Correct answer – B**
27. **Correct answer – B**

Previous Year Questions

1. **Correct answer – D**

   This identifies the specific locations/functions of the thyroid, adrenal, pituitary, and pineal glands.

2. **Correct answer – C**

   This is a mismatch; Calcitonin is produced by the thyroid, while the pancreas produces Insulin/Glucagon.

3. **Correct answer – D**

   Ovaries secrete estrogen and progesterone, but also produce relaxin and inhibin during different cycles.

4. **Correct answer – C**

   If Y represents the pancreas (as is common in these diagrams), its deficiency leads to diabetes, not goitre.

5. **Correct answer – B**

   This is the standard matching for metabolic, stress, sexual, and sugar-regulating hormones.

6. **Correct answer – D**

   Karan has goitre (thyroid/Q) and Ali has diabetes (pancreas/R), making both statements correct.

7. **Correct answer – C**

   Pituitary secretes growth hormones/tropic hormones, while parathyroid (not mentioned in that specific context) handles calcium.

8. **Correct answer – A**

   Only the statement about Parathyroid hormone regulating calcium and phosphate is true.

5 Conservation of Plants and Animals

Solutions

Worksheet – 1

1. **Correct answer – A**

   While the term was originated by others, Edward O. Wilson popularized it to describe the immense variety of life on Earth.

2. **Correct answer – C**

   Genetic diversity allows populations to adapt to changes, such as diseases or climate shifts, ensuring their long-term survival.

3. **Correct answer – B**

   Over 50,000 varieties of rice and nearly 1,000 varieties of mangoes. This illustrates

   diversity at the genetic level within single species found throughout the Indian subcontinent.

4. **Correct answer – D**

   This is a straightforward measure of how many different species are present in a specific geographical location.

5. **Correct answer – C**

   The Western Ghats are a biodiversity hotspot with a much higher concentration of unique frog and toad species than the Eastern Ghats.

6. **Correct answer – D**

   Ecological diversity focuses on the variety of habitats, communities, and biological processes in the biosphere.

7. **Correct answer – C**

   India's vast range of climates and terrains creates more diverse habitats compared to the relatively uniform landscape of Scandinavia.

8. **Correct answer – A**
   This figure represents the species that have been scientifically named and recorded, though millions more likely remain undiscovered.

9. **Correct answer – C**
   Out of every ten animals on the planet, approximately seven are insects, making them the most species-rich taxonomic group

10. **Correct answer – B**
    Many microbes cannot be grown in labs, and their biochemical differences make them harder to classify than plants or animals.

worksheet – 2

1. **Correct answer – C**
   This is considered the most significant factor because it destroys the homes and resources species need to survive.

2. **Correct answer – B**
   The Amazon rainforest Often called the "lungs of the planet," vast tracts are being cleared specifically for agriculture and industrial cattle ranching.

3. **Correct answer – A**
   Over-Exploitation These species were hunted or harvested by humans at a rate far faster than they could naturally replenish their populations.

4. **Correct answer – D**
   Nile perch the introduction of this large predatory fish into Lake Victoria disrupted the entire ecosystem, leading to a massive loss of native fish species.

5. **Correct answer – B**
   Co-Extinctions This occurs when a species that is dependent on another (like a parasite on a specific host) disappears because its partner has gone extinct.

6. **Correct answer – C**

   Great Indian bustard, Gharial, Javan rhinoceros These species face an extremely high risk of extinction in the wild and require immediate, intensive conservation efforts.

7. **Correct answer – B**

   1964 Since its establishment, the Red List has become the world's most comprehensive source of information on the global conservation status of species.

8. **Correct answer – A**

   Bioprospecting This involves the systematic search for biochemical and genetic information in nature that can be developed into commercial products like medicine.

9. **Correct answer – C**

   Broadly Utilitarian This argument focuses on the "ecosystem services" that nature provides for free, such as oxygen production, pollination, and climate regulation.

10. **Correct answer – D**

    Ethical Arguments This perspective holds that every species has an intrinsic right to exist, regardless of whether it provides any direct benefit to humans.

worksheet 3

1. **Correct answer – C**

   Biosphere Reserve: This protects the entire ecosystem and its inhabitants in their natural home, unlike seed banks or gardens.

2. **Correct answer – B**

   36: There are currently 36 recognized biodiversity hotspots globally, characterized by high levels of species richness and endemism.

3. **Correct answer – C**

   The Transition zone is the outermost area where local communities and conservationists work together for sustainable human activities.

4. **Correct answer – B**

   They are species found only in a specific region and nowhere else: Endemic species are unique to a particular geographical location, such as a specific island or mountain range.

5. **Correct answer – D**

   Aravalli hills: The Aravalli hills in Rajasthan are home to sacred groves that act as important refuges for native flora and fauna.

6. **Correct answer – C**

   Wildlife Sanctuaries: According to the data, sanctuaries cover the largest total land area among protected categories in India.

7. **Correct answer – B**

   Cryopreservation: This involves storing biological material at extremely low temperatures (usually in liquid nitrogen) to keep them viable indefinitely.

8. **Correct answer – C**

   To address biological diversity and global environmental integrity: This historic summit led to the Convention on Biological Diversity (CBD) to promote sustainable development.

9. **Correct answer – A**

   CITES stands for the Convention on International Trade in Endangered Species. It ensures that global trade doesn't drive wild species toward extinction.

10. **Correct answer – C**

    National Parks are "strictly reserved" while Sanctuaries allow "controlled biotic interference." National Parks provide the highest level of protection, whereas sanctuaries allow limited human interaction like grazing.

Competitive Corner

SINGLE OPTION CORRECT TYPE

1. **Correct answer – B**
   2.4%: India is highly efficient in biodiversity, supporting a huge percentage of species on a very small fraction of the world's land.
2. **Correct answer – A**
   Higher productivity and less year-to-year variation in biomass: Tilman's experiments proved that more diverse plots are more stable and produce more biomass over time.
3. **Correct answer – C**
   Decrease in plant production (productivity): Losing species weakens the ecosystem's overall ability to capture energy and produce food.
4. **Correct answer – B**
   ***Clarias bactrachus*:** The invasive African catfish competes with and threatens the survival of our native Indian catfish.
5. **Correct answer – D**
   Nile Perch: While the Nile Perch is a dangerous invasive species, it is a fish, not a "weedy plant" like Lantana or Parthenium.
6. **Correct answer – C**
   Snow leopard: The snow leopard is listed as "Vulnerable" or "Threatened," whereas the others are in the most critical danger category.
7. **Correct answer – B**
   More than 25%: A quarter of all modern medicines are derived directly from chemicals found in plants.
8. **Correct answer – D**
   Assam: Manas is a famous Biosphere Reserve and UNESCO World Heritage site located in the Himalayan foothills of Assam.
9. **Correct answer – B**
   Mandumalai Wildlife Sanctuary: Based on historical records in textbook tables, Mudumalai (Tamil Nadu) is one of India's oldest sanctuaries.
10. **Correct answer – C**
    Kanger Ghati National Park: This park is located in the Bastar region of Chhattisgarh and is known for its limestone caves.

ONE OR MORE THAN ONE OPTION CORRECT TYPE

11. **Correct answer – A, B, C, D**

    Deforestation is a "quadruple threat" that ruins weather patterns, soil quality, animal homes, and global temperatures.
12. **Correct answer – B, C, D**

    A stable community acts like a healthy body; it stays consistent and fights off "infections" like invasive species.
13. **Correct answer – A, B, D**

    Rainforests have shrunk from 14% to 6% of land area, and breaking them into fragments hurts large-roaming animals the most.
14. **Correct answer – B, C, D**

    These are "narrow" because they provide direct, sellable goods like wood and medicine, whereas flood control is a "broad" service.
15. **Correct answer – A, B, C, E**

    The transition zone is for human use, the buffer for research/education, and the core is strictly off-limits to everyone.

ASSERTION AND REASNING TYPE

16. **Correct answer – D**
    We cannot identify all microbes because most "refuse" to grow in labs for us to study.
17. **Correct answer – A**

    Both are true and R explains why habitat loss (specifically fragmentation) leads to survival difficulties.
18. **Correct answer – C**

    India is a mega-diversity country because it supports high global diversity (8.1%) on little land (2.4%).

19. **Correct answer – A**
The strict boundaries and rules of National Parks are designed to give wildlife total priority.

MATCH THE FOLLOWING TYPE

20. **Correct answer – B**
a-Q, b-R, c-P
21. **Correct answer – D**
a-Q, b-S, c-P, d-R

COMPREHENSHION TYPE

Passage1

22. **Correct answer – B**
Stability means the ecosystem can "bounce back" from problems and keep working regularly.
23. **Correct answer – A**
Paul Ehrlich used the plane to show that even "small" parts are vital for the whole system to stay "in the air."
24. **Correct answer – C**
If you pull out enough rivets (species), the whole plane (ecosystem) will eventually crash.

Passage2

25. **Correct answer – B**
Two: These are *In-situ* (protecting the home) and *Ex-situ* (protecting the individual elsewhere).
26. **Correct answer – C**
Conservation within natural ecosystems: It is the practice of protecting a species right where it lives in the wild.
27. **Correct answer – D**
National Park: National Parks protect the natural landscape, whereas the other options are "off-site" or lab-based methods.

Previous Year Questions

1. **Correct answer – A**
Zoological parks are an ex-situ conservation method because they protect components of biological diversity outside their natural habitats.
2. **Correct answer – D**
The statement is incorrect because a gene bank, rather than an arboretum, is the collection responsible for maintaining seeds, tissue culture, and frozen germplasm.
3. **Correct answer – D**
This match is incorrect because Gir National Park is located in Gujarat, not Jharkhand.
4. **Correct answer – C**
Kaziranga National Park is famous for the one-horned rhinoceros, and the Sundarbans is the well-known habitat of the Royal Bengal Tiger.
5. **Correct answer – D**
All statements are correct because species characteristics should be common, and the paragraph swaps the definitions of endemic (confined to one area) and exotic (introduced) species.
6. **Correct answer – C**
The transition (or manipulation) zone is the outermost part of a biosphere reserve where human activities like cropping and forestry are permitted.
7. **Correct answer – A**
Statement (iii) is true because a decrease in trees leads to floods, while statement (iv) is false because planting trees on previously non-forested land is called afforestation.
8. **Correct answer – B**
In the provided graph, curve R shows a species that becomes extinct by year 20 (like the Passenger pigeon), while curve P shows a population recovering through conservation.
9. **Correct answer – C**

Water hyacinth is a classic example of an exotic species introduced into a new geographical area.

10. **Correct answer – C**

Deforestation, pollution, and the introduction of exotic species are significant human-driven threats that lead to the loss of biodiversity.

6 CROP PRODUCTION AND MANAGEMENT

Solutions

Worksheet – 1

1. **Correct Answer (C) Kharif crops**

   Kharif crops are also known as monsoon crops because they are sown at the beginning of the rainy season (June–July) and harvested in autumn.

   Rabi crops are grown in winter, Zaid crops are grown in summer, and cash crops are grown for profit.

2. **Correct answer (C) Zaid crops**

   Zaid crops are short-duration crops grown during the summer months (March–June) between the Rabi and Kharif seasons. Watermelon, muskmelon, and cucumber are classic examples.

3. **Correct Answer (C) Irrigation**

   The process of artificially supplying water to land or crops to help growth is called irrigation. Harvesting is gathering crops, storage is keeping produce safe, and sowing is planting seeds.

4. **Correct Answer (B) Weeding**
   Weeding is the removal of unwanted plants (weeds) that compete with the main crop for nutrients, sunlight, and water. Ploughing prepares the soil.

5. **Correct Answer (C) Human or animal power**
   Traditional irrigation methods like the moat (pulley system), chain pump, dhekli, and rahat rely on manual labor or livestock rather than modern technology like electricity or solar panels.

6. **Correct Answer (B) Irrigation**

   In areas with irregular or insufficient rainfall, irrigation becomes the most crucial operation to ensure crops get a steady water supply for survival.

7. **Correct Answer (B) Protect produce from pests and microorganisms**

   Freshly harvested grains contain moisture, making them prone to fungus and pests. Storage in granaries or silos protects them from damage.

8. **Correct Answer (C) Ploughing**

   Ploughing (or tilling) loosens and turns the soil, which improves aeration, allowing roots to breathe more easily and helping beneficial microbes grow.

9. **Correct Answer (A) Proper seed spacing during sowing**

   Sowing seeds at the correct distance prevents overcrowding, ensuring each plant has enough space to access sunlight, water, and nutrients without competing with neighbors.

10. **Correct Answer (C) October–November**

    Rabi crops are winter crops typically sown at the beginning of winter (October– November) and harvested in the spring (March–April).

Worksheet-2

1. **Correct Answer (C) Simple selection method**

   Jai Prakash Singh is a renowned farmer-innovator from Varanasi who developed several high-yielding varieties (HYVs) of wheat, paddy, and mustard by identifying and selecting superior plants from his own fields.

2. **Correct Answer (C) Development of HYVs and DRVs**

   The Green Revolution centered on the use of High-Yielding Varieties (HYVs) and Disease-Resistant Varieties (DRVs), combined with modern irrigation and chemical fertilizers.

3. **Correct Answer (B) Breed quality**

   While feeding and care are vital, the potential milk yield of an animal is primarily determined by its genetic makeup or breed quality.

4. **Correct Answer (B) Genetic variability**

   Genetic variability is the root of all breeding programs; without differences in traits like yield or resistance, there is nothing for breeders to select from.

5. **Correct Answer (D) Soil ploughing**

   Soil ploughing is a crop production activity. While breed, feed, and housing (ventilation) affect cattle health and milk, soil preparation does not.

6. **Correct answer(A) Determines the traits passed to hybrids**

   Selecting the best parents ensures that desirable characteristics, such as high yield or pest resistance, are combined in the resulting hybrid offspring.

7. **Correct Answer (D) 40 litres/day**

   The Holstein-Friesian is the highest-producing dairy breed in the world; high-performing individuals can easily exceed 40 litres per day under optimal

8. **Correct Answer (C) Cultivable land is limited**

   Since the total area of agricultural land is nearly fixed, we must increase the yield per unit area (productivity) to feed a growing population.

9. **Correct Answer (C) Uniformity and trait stability**

   Repeated self-pollination leads to homozygosity, which ensures that the offspring consistently show the same superior traits as the parents (true-breeding).

10. **Correct Answer (C) Prevent diseases and maintain health**

    Regular check-ups by a veterinarian are essential to identify illnesses early,

    administer vaccinations, and ensure the livestock remains productive.

Worksheet-3

1. **Correct Answer (B) Maintain and improve desired traits**

   Inbreeding involves mating closely related individuals to increase homozygosity, which helps in evolving pure lines and maintaining desirable traits within a breed.

2. **Correct Answer (B) Have below-average productivity**

   Out-crossing is the best method for animals that are below average in productivity (milk yield, growth rate, etc.) as a single out-cross often helps overcome inbreeding depression.

3. **Correct Answer (C) Crossbreeding**

   Hisardale is a famous breed of sheep developed in Punjab by crossbreeding Bikaneri ewes and Marino rams.

4. **Correct Answer (C) Interspecific hybridization**

   A mule is produced by crossing two different related species—a male donkey and a female horse—which is called interspecific hybridization.

5. **Correct Answer (B) Layers**

   In poultry farming, birds raised for egg production are called layers, while those raised for meat are called broilers.

6. **Correct Answer (C) Unfertilized eggs**

   In a bee colony, drones (males) are produced from unfertilized eggs through a process called parthenogenesis.

7. **Correct Answer (B) Communication of nectar location**

   Honey bees use the waggle dance to communicate the direction and distance of a food source (nectar/pollen) to other members of the hive.

8. **Correct answer (C) Queen**

   While all bee larvae are fed royal jelly initially, only the larva chosen to become the Queen is fed royal jelly throughout her entire development.

9. **Correct Answer (C) 4–6 generations**
Out-crossing is defined as mating animals of the same breed that have no common ancestors on either side of their pedigree for 4–6 generations.
10. **Correct Answer (B) High biological quality of protein**
The Protein Efficiency Ratio (PER) measures the effectiveness of a protein source. Eggs have a high PER because they contain all essential amino acids in the right proportions, indicating high biological value.

Competitive Corner

SINGLE OPTION CORRECT TYPE

1. **Correct Answer (B) To ensure uniform seed placement and smooth irrigation.**
Levelling prevents soil erosion by wind/water and ensures that water and seeds are distributed evenly across the field.
2. **Correct Answer (C) Interspecific hybridization**
As seen in your previous worksheet, a mule is the result of crossing a male donkey and a female horse, which are two different species.
3. **Correct Answer (B) Pisciculture refers exclusively to the culture of finned fishes.**
Aquaculture is a broad term for farming all aquatic organisms (shrimp, oysters, seaweed), while pisciculture is the specific term for farming fish.
4. **Correct Answer (C) Andhra Pradesh**
Andhra Pradesh is consistently ranked as the largest egg-producing state in India, often referred to as the "Egg Bowl of Asia."
5. **Correct Answer (C) It provides the diverse traits (alleles) that breeders can select from to improve crops.**
Breeders need variation (like differences in height or grain size) to choose and combine the best possible traits into a new variety.
6. **Correct Answer (A) Simple selection involves choosing the best existing plants; hybridization involves manually crossing two different parents.**
Simple selection is about finding the "best of the bunch" already in the field, whereas hybridization is a manual process of cross-pollinating two different varieties.
7. **Correct Answer (C) Harvesting**
A Combine Harvester is a versatile machine designed to harvest, thresh, and winnow grain crops in a single operation.
8. **Correct Answer (C) Hybridization**
The deliberate transfer of pollen from one plant to the stigma of another to create a genetically diverse offspring is called hybridization.
9. **Correct answer (B) Rohu**
Labeo rohita is the scientific name for Rohu, one of the most important Indian major carps used in aquaculture.
10. **Correct answer (B) Cousin mating**
Line breeding is a milder form of inbreeding that involves mating distantly related animals, such as cousins, to maintain the traits of a specific admired ancestor.

ONE OR MORE THAN ONE OPTION CORRECT TYPE

11. **Correct Answer A), B), and C)**
For a profitable fishery, selected organisms should have high disease resistance to survive high-density environments, a fast rate of growth to reach market size quickly, and high nutritional value for consumers.
12. **Correct Answer A), B), and C)**
In a bee colony:
The Queen is the only fertile, diploid female responsible for laying eggs.
Drones are haploid males that develop from unfertilized eggs via parthenogenesis.
Workers are sterile, diploid females that perform all hive tasks like foraging and nursing.

13. **Correct Answer A), C), and D)**

Out-breeding refers to mating unrelated animals and includes three main types: out-crossing (same breed, no common ancestors), crossbreeding (different breeds), and interspecific hybridization (different but related species).

14. **Correct Answer A), B), and C)**

Exotic breeds are foreign breeds imported to India for their superior milk production. Common examples include the Holstein-Friesian (Netherlands), Jersey (Island of Jersey), and Brown Swiss (Switzerland).

15. **Correct Answer A), C), and D)**

Kharif crops are sown at the start of the monsoon season. Rice, maize, and soybean are primary examples of Kharif crops.

ASSERTION AND REASONING TYPE

16. **Correct Answer D) A is false but R is true**

Assertion (A) is False: Inbreeding does not always increase productivity; while it helps in developing pure lines, it can also lead to the expression of harmful recessive genes.

Reason (R) is True: Continuous close inbreeding is well-known to cause inbreeding depression, which actually reduces fertility and productivity.

17. **Correct Answer B) Both A and R are true but R is NOT the correct explanation of A**

Assertion (A) is True: Mules are almost always sterile due to an odd number of chromosomes that prevents normal meiosis.

Reason (R) is True: A mule is indeed the offspring of a male donkey and a female horse.

While the parentage is correct, simply stating the parents doesn't explain the biological mechanism (chromosomal mismatch) of why the offspring is sterile.

18. **Correct Answer A) Both A and R are true and R is the correct explanation of A**

Assertion (A) is True: Proper soil preparation (ploughing and levelling) allows roots to penetrate deeper into the earth.

Reason (R) is True: Ploughing loosens the soil, which directly improves aeration and drainage, making it easier for roots to grow and breathe.

19. **Correct Answer B) Both A and R are true but R is NOT the correct explanation of A**

Assertion (A) is True: Aquaculture is a broad category that includes the farming of crustaceans, molluscs, and aquatic plants in addition to fish.

Reason (R) is True: Pisciculture is specifically the breeding and rearing of finned fishes.

Both are correct definitions, but the fact that pisciculture is limited to fish is not the reason why aquaculture is broad; they are simply two different classifications

MATRIX MATCH TYPE

20. **Correct Answer (A) A–R, B–P, C–Q, D–S**
21. **Correct Answer (A) A–R, B–Q, C–P, D–S**

COMPREHENSION TYPE

Passage-1

22. **Correct Answer (B) Water-scarce regions**
23. **Correct Answer (B) Acts as a solvent for nutrients**
24. **Correct Answer (C) Uniform water distribution**

Passage-2

25. **Correct Answer (C) Meat production**
26. **Correct Answer (B) Protein**
27. **Correct Answer (C) Maintain bird health**

Previous Year Questions

1. **Correct Answer (A) It is rich in inorganic nutrients, i.e., nitrogen, phosphorus and potassium.**

   This is a characteristic of fertilizers, not manure. Manure is organic matter that improves soil texture and water-holding capacity but is relatively low in concentrated inorganic nutrients like NPK.

2. **Correct Answer (B) It is used to sow seeds at an equal distance and proper depth.**

   The implement typically shown in this NSO question is a seed drill. It ensures seeds are placed at the correct depth and distance and then covered with soil to prevent damage from birds.

3. **Correct Answer (B) P could be dhekli whereas R could be drip irrigation.**

   In these flowcharts, P usually represents traditional methods (like dhekli, moat, or Rahat) that are often labor-intensive, while R represents modern methods (like drip irrigation or sprinklers) which are more water-efficient.

4. **Correct Answer (C) Fungi**

   Both Rust (caused by Puccinia) and Smut (caused by Ustilago) are well-known fungal diseases that affect wheat and other cereal crops.

5. **Correct Answer (A) X: Cotton, Y: Mustard**

   Cotton is a Kharif crop sown during the monsoon (June–October), while Mustard is a Rabi crop grown in the winter (November–April).

   Groundnut (Kharif) and Maize (Kharif); Pea (Rabi) and Gram (Rabi); Wheat (Rabi) and Paddy (Kharif).

7 Why do We Fall Ill

Solutions

Worksheet – 1

1. **Correct Answer (C) Health of the whole community**
   Community health focuses on the physical and mental well-being of the people in a specific geographic region or group.
2. **Correct Answer(B) Discomfort or lack of ease**
   The word "disease" literally breaks down into "dis-ease," meaning a departure from a state of ease or comfort.
3. **Correct Answer (C) Diabetes**
   Chronic diseases are long-term conditions that usually progress slowly and last for months or years, such as diabetes or cardiovascular diseases.
4. **Correct Answer (C) Common cold**
   Acute diseases are characterized by a rapid onset and short duration, like the common cold or flu.
5. **Correct Answer (B) Bacteria**
   Tuberculosis (TB) is an infectious disease caused by the bacterium Mycobacterium tuberculosis.
6. **Correct Answer (D) Physical, mental and social well-being**
   According to the World Health Organization (WHO), health is a state of complete physical, mental, and social well-being, not just the absence of disease.
7. **Correct Answer (A) Virus**
   Yellow fever is an acute viral haemorrhagic disease transmitted by infected mosquitoes.
8. **Correct Answer (A) Ringworm**
   Ringworm is a common fungal infection of the skin.
9. **Correct Answer (A) Typhoid**
   Typhoid is a waterborne disease caused by the bacterium Salmonella typhi, often spread through contaminated water or food.
10. **Correct Answer (C) Pathogens**
    Infectious diseases are caused by harmful microorganisms called pathogens, which include bacteria, viruses, fungi, and parasites.

Worksheet-2

1. **Correct Answer(B) Congenital disorder**
   Colour blindness is a condition typically present from birth due to genetic mutations. While it is primarily classified as a hereditary disease, in the context of these options, it is a congenital disorder (present at birth).
2. **Correct Answer (B) Non-hereditary**
   Acquired diseases are those that develop after birth. Unlike hereditary diseases, they are not passed down through genes from parents.
3. **Correct Answer (B) Infectious agents**
   Communicable diseases are caused by pathogens (infectious agents) like bacteria, viruses, fungi, or parasites that can spread from one person to another.
4. **Correct Answer (C) Vitamin C**
   Scurvy is a classic deficiency disease specifically caused by a lack of Vitamin C in the diet.
5. **Correct Answer (B) Hereditary disease**
   A hereditary disease is a condition passed from parents to offspring through defective genes or DNA abnormalities.

6. **Correct Answer (C) Non-hereditary diseases**
   Since acquired diseases are developed during a person's lifetime and are not part of their genetic makeup from birth, they are classified as non-hereditary.
7. **Correct answer (C) Non-infectious diseases**
   Non-communicable diseases (NCDs) are those that cannot be spread from person to person through contact or pathogens, making them non-infectious.
8. **Correct Answer (C) Virus**
   A virus is an infectious agent that causes communicable diseases (like the flu or COVID-19).
9. **Correct Answer (B) Non-communicable**
   A disease caused by a nutrient deficiency (like vitamins or minerals) stays within the individual and cannot be transmitted to others.
10. **Correct Answer (B) Present at birth**
    Congenital diseases are medical conditions that are present from the time of birth, regardless of whether they are hereditary or caused by environmental factors during pregnancy.

Worksheet-3

1. **Correct Answer (C) Typhoid**
   Typhoid is caused by the bacterium Salmonella typhi.
2. **Correct Answer (B) Helminths**
   Elephantiasis (Filariasis) is caused by parasitic filarial worms, which are a type of helminth (roundworm).
3. **Correct Answer (D) Fungi**
   Ringworm is a common skin infection caused by a fungus, not a worm.
4. **Correct Answer (A) Swelling and redness**
   Inflammation is the body's immune response to injury or infection, typically marked by redness, swelling, heat, and pain.
5. **Correct Answer (C) Antiviral drug**
   Oseltamivir (Tamiflu) is a medication used to treat and prevent influenza (flu) viruses.
6. **Correct Answer (C) Antibodies**
   A vaccine trains the immune system to recognize pathogens by stimulating the production of antibodies.
7. **Correct Answer (C) Do not spread from person to person**
   Non-communicable diseases (NCDs) are non-infectious conditions that cannot be transmitted between people.
8. **Correct Answer (D) Antiprotozoal drugs**
   Diseases caused by protozoa (like malaria or amoebiasis) are treated with specific antiprotozoal medications.
9. **Correct Answer (A) Lungs**
   Tuberculosis (TB) most commonly attacks the lungs (pulmonary TB), though it can affect other parts of the body.
10. **Correct Answer (D) Chronic**
    Most non-communicable diseases, such as diabetes or heart disease, are chronic because they persist for a long duration.

COMPETITIVE CORNER

SINGLE OPTION CORRECT TYPE

1. **Correct Answer (C) Proper disposal of sewage and town wastes.**
   Community health focuses on environmental and public systems managed by local authorities to benefit everyone, such as waste management.
2. **Correct Answer (C) Prolonged loss of work efficiency and long-lasting effects.**
   Chronic diseases last for a long time, causing significant long-term impact on health and productivity.
3. **Correct Answer (B) Lungs and respiratory tract**
   Pathogens entering through the nose typically follow the respiratory pathway, targeting the lungs or nasal passages.
4. **Correct Answer (A) Typhus Fever**
   Typhus fever is caused by bacteria of the genus Rickettsia, usually spread by fleas or lice.
5. **Correct Answer (B) Paracetamol**
   Paracetamol is a widely used analgesic (painkiller) and antipyretic (fever reducer).

6. **Correct Answer (B) Droplet infection through the air**

   Sneezing releases tiny droplets that can carry pathogens through the air to others nearby.

7. **Correct Answer (B) Because the genetic instruction for the disease is present from the time of birth**

   Congenital simply means "present at birth." Since hereditary diseases are written in the DNA at conception, they are present when the person is born.

8. **Correct Answer (A) To see if the headache is a symptom of a more serious structural change, like meningitis.**

   Doctors use tests to distinguish between simple symptoms and serious underlying diseases that may manifest as a headache.

9. **Correct Answer (B) Fungus**

   Athlete's foot (Tinea pedis) is a fungal infection that usually starts between the toes.

10. **Correct Answer (B) Acquired disease**

    Any disease that develops after birth due to environmental factors, infections, or lifestyle is called an acquired disease.

ONE OR MORE THAN ONE OPTION CORRECT TYPE

11. **Correct Option A) Redness, B) Pain, C) Swelling**

    Inflammation is characterized by the five cardinal signs: redness (rubor), swelling (tumor), heat (calor), pain (dolor), and loss of function.

12. **Correct option A) Diabetes, B) Hypertension, D) Arthritis**

    Diabetes, Hypertension (high blood pressure), and Arthritis are non-communicable diseases because they are not caused by infectious agents and cannot be passed from person to person.

13. **Correct option A) Cuts or wounds, B) Respiratory tract, C) Insect bites**

    Pathogens enter the body through various portals, including breaks in the skin (cuts/wounds), the respiratory tract (breathing in droplets), and vectors (insect bites).

14. **Correct option A) It stimulates the formation of antibodies, B) It is a suspension of killed or weakened microorganisms, C) The antibodies produced remain in the blood to destroy future germs**

    Vaccines use weakened/killed microbes to trigger the immune system to create antibodies and memory cells that provide long-term protection.

15. **Correct option A) Scurvy, B) Anaemia, C) Goitre**

    Deficiency diseases result from a lack of specific nutrients: Scurvy (Vitamin C), Anaemia (Iron), and Goitre (Iodine).

ASSERTION AND REASONING TYPE

16. **Correct Option (A) Both A and R are true, and R is the correct explanation of A.**

    A vector is defined as an organism that carries and transmits an infectious pathogen into another living organism. Mosquitoes do exactly this by carrying pathogens like those for Malaria or Dengue between hosts.

17. **Correct option (D) A is false, but R is true.**

    Assertion (A) is false because Colour blindness is a **congenital/hereditary** disease, not an acquired one. Reason (R) is true as it correctly describes how the condition is inherited through genes.

18. **Correct option (D) A is false, but R is true.**

    Assertion (A) is false because inflammation is actually a defense mechanism showing the immune system is actively fighting the pathogen, not that it is being defeated. Reason (R) is a true statement describing the symptoms.

19. **Correct option (B) Both A and R are true, but R is NOT the correct explanation of A.**
Both statements are scientifically and socially true. However, the Reason (R) discusses environmental sanitation, which does not explain why stable income helps a person afford food (Assertion A). There is no logical link between the two.

MATRIX MATCH TYPE

20. **Correct Option (A) a–Q, b–P, c–R, d–S**
21. **Correct Option (C) a–S, b–R, c–P, d–Q**

COMPREHENSION TYPE

22. **Correct option (B) Contaminated water**
23. **Correct option (C) Non-infectious disease**
24. **Correct option (B) Malaria**
25. **Correct option (C) Spread of infectious diseases**
26. **Correct option (B) Weakened or killed pathogens**
27. **Correct option (B) Destroy pathogens**

PREVIOUS YEAR QUESTIONS

1. **Correct option(A) They last for a long time, may be as much as lifetime**
This statement describes chronic diseases. Acute diseases are defined by their short duration and rapid recovery.
2. **Correct option(D) Osteomalacia**
Osteomalacia is the softening of bones in adults specifically caused by a severe Vitamin D deficiency.
3. **Correct Answer (C) Cleft palate is caused due to environmental factors whereas Haemophilia is caused due to gene mutation.**
Cleft palate can be influenced by environmental factors (like maternal health), while Haemophilia is a classic genetic disorder caused by a gene mutation.
4. **Correct Answer (C) Chicken pox — Bacteria**
This is the incorrect match. Chickenpox is caused by the Varicella-zoster virus, not bacteria.
5. **Correct Answer (C) T — F — T — T**

8 Fundamentals of ECOLOGY

Solutions

Worksheet – 1

1. **Correct Answer: C) Biosphere**
   The Biosphere is the global sum of all ecosystems where living organisms interact with the lithosphere, hydrosphere, and atmosphere to recycle nutrients and energy.
2. **Correct Answer: B) A blanket of gases surrounding a planet**
   Scientific definitions describe the atmosphere as a gaseous envelope held to the Earth by gravity.
3. **Correct Answer: C) Stratosphere**
   Approximately 90% of the Earth's ozone is concentrated in the Stratosphere, specifically in its lower portion.
4. **Correct Answer: B) Mesosphere**
   Gases in the Mesosphere are dense enough to create friction that causes meteors to burn up.
5. **Correct Answer: C) Interaction between living organisms**
   "Biotic" relates to all living entities, including their interactions like predation, competition, and symbiosis.
6. **Correct Answer: B) Altitude and mountain direction**
   Physiographic factors are associated with the physical nature of an area, such as slopes, altitude, and landforms.
7. **Correct Answer: A) 10,000 kilometres**
   Most standard definitions place the upper limit of the Exosphere at about 10,000 km, where it merges with solar wind.
8. **Correct Answer: A) Soil formation and properties**
   The term "edaphic" specifically refers to soil-related abiotic factors like pH, texture, minerals, and moisture.
9. **Correct Answer: D) Thermosphere**
   Auroras (Northern and Southern Lights) occur in the Thermosphere when charged particles from the sun collide with atmospheric gases.
10. **Correct Answer: C) Edaphic and physiographic factors**
    While factors like light and temperature act directly on organisms, edaphic (soil) and physiographic (landform) factors are often considered indirect because they influence the direct factors (like water availability or temperature exposure).

WORKSHEET-2

1. **Correct Answer: D) Living organisms interacting with physical environment**
   An ecosystem consists of all the living (biotic) communities in a given area interacting with their non-living (abiotic) physical environment.
2. **Correct Answer: C) Succession**
   Ecological succession is the process by which the structure of a biological community evolves over time, moving from pioneer species to a stable, complex climax community.
3. **Correct Answer: C) A. G. Tansley**
   British botanist Arthur Tansley coined the term in 1935 to describe the interactive system between biotic and abiotic components.
4. **Correct Answer: B) Developed and maintained by humans**
   Artificial ecosystems, like croplands, gardens, or aquariums, require human intervention for their survival and management.

5. **Correct Answer: D) Use sunlight to make food**
   "Photo" means light and "autotroph" means self-feeding; photoautotrophs convert light energy into chemical energy (glucose) via photosynthesis.
6. **Correct Answer: D) Primary consumers**
   Both grasshoppers and rabbits are herbivores, meaning they eat producers (plants) directly, making them the first consumer level.
7. **Correct Answer: B) Energy flow and material cycling**
   The fundamental operation of any ecosystem is the movement of energy (from the sun through food chains) and the cycling of nutrients (like carbon and nitrogen).
8. **Correct Answer: A) Occurs through different trophic levels**
   Energy enters through producers and flows unidirectionally to various trophic levels (primary, secondary, and tertiary consumers).
9. **Secondary consumers mainly feed on:**
   C) Primary consumers
   Secondary consumers are carnivores or omnivores that eat primary consumers (herbivores).
10. **Correct Answer: C) Decomposers**
    Fungi like mushrooms and Rhizopus (bread Mold) are saprotrophs that break down organic matter from dead plants and animals.

WORKSHEET-3

1. **Correct Answer: C) Transfer of energy from one trophic level to another**

   A food chain depicts the linear sequence of organisms through which nutrients and energy pass as one eats another.

2. **Correct Answer: B) Secondary consumer**

   In a typical three-step chain, the lion eats the deer (primary consumer), making it a secondary consumer or carnivore.

3. **Correct Answer: D) Primary consumers**

   Deer and cows are herbivores that feed directly on producers (grass), placing them at the second trophic level as primary consumers.

4. **Correct Answer: B) Tertiary consumers**

   Apex predators or scavengers like hawks and vultures often occupy the top of the food chain, feeding on secondary consumers.

5. **Correct Answer: A) Organisms die**

   The detritus food chain starts with dead organic matter, such as animal remains or fallen leaves, which are then consumed by microorganisms and detritivores.

6. **Correct Answer: A) They increase ecosystem stability**

   Food webs offer multiple feeding pathways; if one species fails, others can fill the role, ensuring ecosystem stability.

7. **Correct Answer: A) Unidirectional**

   Energy flows in one direction (from sun to producers to consumers) and is lost as heat at each step; it cannot flow back to the previous level.

8. **Correct Answer: C) Greater food alternatives**

   A stable ecosystem has complex food webs that provide organisms with multiple food sources, making the system resilient to change.

9. **Correct Answer: B) Animal biomass**

   When a deer (consumer) eats plants, it uses the nutrients to build its own tissues, effectively converting plant energy into animal biomass.

10. **Correct Answer: A) Sunlight**

    Grass is an autotroph (producer) that captures solar energy through photosynthesis to create its own food.

WORKSHEET-4

1. **Correct Answer: B) Producers**
   All ecological pyramids start with producers (autotrophs) at the bottom because they are the foundation of energy for all other trophic levels.
2. **Correct Answer: C) Upright**
   According to the 10% law, energy is lost as heat at each transfer, so the pyramid of energy can never be inverted or irregular.
3. **Correct Answer: D) Nudation**
   Nudation is the first stage of ecological succession, referring to the formation of a bare area due to factors like floods, erosion, or volcanic eruptions.
4. **Correct Answer: B) Climax community**
   A climax community is the final, stable stage of succession that remains in equilibrium with the local climate.
5. **Correct Answer: C) Commensalism**
   In commensalism, one species benefits (remora gets a ride and food scraps) while the other (shark) is neither helped nor harmed.
6. **Correct Answer: B) Predator eats prey**
   This is a direct interaction where the predator kills and consumes the prey for energy.
7. **Correct Answer: C) Spindle-shaped**
   A single large producer (one tree) supports many herbivorous birds, which are then eaten by a few hawks. This creates a base that is narrow, a middle that is wide, and a top that is narrow—a spindle shape.
8. **Correct Answer: A) Soil is already present**
   Since secondary succession occurs in areas where a community previously existed, the soil and seeds are often already there, skipping the long process of soil formation.
9. **Correct Answer: C) Parasitism**

   A tapeworm is a parasite that benefits by taking nutrients from the host (human), who is harmed in the process.
10. **Correct Answer: B) Ammensalism**

    This is a specific type of ammensalism (antibiosis) where the bacteria are killed or inhibited while the Penicillium remains unaffected.

Competitive Corner

SINGLE OPTION CORRECT TYPE

1. **Correct Answer: B) Physiographic; Climatic**

   Physiographic factors (like mountain range direction) directly influence climatic factors (like wind patterns and rainfall).
2. **Correct Answer: B) It is Secondary Succession starting from a previously built-up substrate.**

   Secondary succession is faster because the soil and nutrient base already exist, unlike primary succession on bare rock or lava.
3. **Correct Answer: B) It will be inverted at the base because the number of producers is lower than consumers.**

   A single tree (one producer) supports thousands of insects (many consumers), making the base of the numerical pyramid very narrow or inverted.
4. **Correct Answer: C) Predators of the Grasshoppers will seek alternative primary consumers.**

   In a stable food web, predators have multiple prey options. If one prey source (grasshoppers) declines, they switch to others.
5. **Correct Answer: C) Autogenic Succession**

   Autogenic succession is driven by the biotic components within the community itself changing their own environment.
6. **Correct Answer: C) 90%**

   Lindeman's 10% Law states that only 10% of energy is transferred to the next level, meaning 90% is lost as heat or through metabolic processes.

7. **Correct Answer: B) It helps in the breakdown of organic matter and loosens the soil.**

   Earthworms improve soil fertility by decomposing waste and aerating the soil through burrowing.

8. **Correct Answer: B) Land, air, light, and water**

   These represent the four fundamental abiotic components (lithosphere, atmosphere, solar energy, and hydrosphere) that sustain life.

9. **Correct Answer: A) Charles Elton**

   Charles Elton first developed the concept of ecological pyramids, which is why they are often called Eltonian pyramids.

10. **Correct Answer: A) Chemoautotrophs**

    Chemoautotrophs synthesize food using energy from chemical reactions rather than light.

ONE OR MORE THAN ONE OPTION TYPE

11. **Correct Answers: A, B, C, and D**

    All four statements are biologically accurate descriptions of the roles in this specific food chain:

    A & B: Grass (Autotroph) produces the energy; Deer (Herbivore) eats it first.

    C: The Lion eats the primary consumer, making it the secondary consumer.

    D: Bacteria break down the remains of all these organisms, acting as the decomposers.

12. **Correct Answers: A, B, and D**

    While the question asks what occurs "first" (which usually points to energy flow), in an MSQ context, all three are valid consequences:
    A: Sunlight can no longer be converted into chemical energy, so energy flow stops.
    B: Without a base, consumers will starve as their food source disappears.
    D: The entire trophic structure collapses because the foundation is gone.

13. **Correct Answers: A, B, and C**

    Inverted biomass pyramids are unique to aquatic systems for these reasons:

    A & B: Phytoplankton (producers) have a short lifespan and reproduce incredibly fast. They are eaten as soon as they are born.

    C: Because of that fast turnover, the standing crop (biomass) of the consumers (fish/zooplankton) at any single moment is physically larger than the biomass of the producers.

14. **Correct Answers: A, B, and D**

    A: Since soil already exists, the ecosystem doesn't have to wait centuries for rocks to break down.

    B: Seeds, roots, and underground spores often survive the disturbance (like a fire), allowing for quick regrowth.

    D: The nutrient cycle is already functional because the soil is already enriched with organic matter.

15. **Correct Answers: B and D**

    B & D: A high turnover rate and short lifespan of producers allow them to support a larger mass of consumers. Even though there are fewer producers right now, they produce enough total energy over time to feed the larger consumer group.

ASSERTION AND REASONING TYPE

16. **Correct Answer: A) Both A and R are true and R is the correct explanation of A**

    Energy flow is indeed a fundamental function of an ecosystem because it is the process that connects all biotic components. The reason it is considered a "flow" or "function" is specifically because energy moves sequentially through the various trophic levels.

17. **Corerct AnswerD) A is false but R is true**

    The Assertion is false because the pyramid of numbers can be inverted (e.g., many insects feeding on a single large tree). The Reason is technically true in most typical ecosystems (like grasslands), but because the Assertion is a "universal" statement ("always upright"), the entire Assertion fails.

18. **Correct Answer D) A is false but R is true**

    The Assertion is false because weather changes occur in the troposphere, not the mesosphere. The Reason is true; the troposphere is the densest layer and contains about 75-80% of the atmosphere's mass, which is why weather happens there.

19. **Correct Answer: A) Both A and R are true and R is the correct explanation of A**

    A climax community is defined by its stability and equilibrium with the environment. It achieves this stability precisely because it is the final, terminal stage of ecological succession where the species composition no longer changes drastically.

MATRIX MATCH TYPE

20. **Correct Answer (A) a–R, b–P, c–S, d–Q**
21. **Correct Answer (A) a–Q, b–R, c–P, d–S**

COMPREHENSION TYPE

22. **Correct Answer (C) Climax community**
23. **Correct Answer (B) Bare rock**
24. **Correct Answer (B) Biodiversity**
25. **Correct Answer (C) Producers**
26. **Correct Answer (B) Decreasing energy at higher levels**
27. **Correct Answer (C) Lindeman**

9 Pollution of Air and Water

Solutions

Worksheet – 1

1. **Correct Answer (C) Industrial Revolution**
   Coal was the primary fuel source that powered the steam engines and factories during the Industrial Revolution.
2. **Correct Answer (C) Perpetual or inexhaustible resources**
   Renewable resources are those that replenish naturally over time and are often referred to as perpetual resources.
3. **Correct Answer (C) Sun**
   The Sun provides the energy for photosynthesis, weather patterns, and almost all life cycles on Earth. Water, coal, and soil are resources, but their energy typically originates from solar processes or geological transformations driven by heat.
4. **Correct Answer (B) Exhaustible resources**
   Non-renewable resources exist in fixed amounts and cannot be replaced once used up, making them "exhaustible."
5. **Correct Answer (B) Maintain oxygen–carbon dioxide balance**
   Forests absorb carbon dioxide and release oxygen through photosynthesis, acting like a respiratory system for the planet.
6. **Correct Answer (B) Controlling growth and management of forests**
   Silviculture is the science and practice of controlling the establishment, growth, composition, health, and quality of forests to meet diverse needs.
7. **Correct Answer (A) Millions of years to form**
   Fossil fuels like coal and oil take millions of years of heat and pressure to develop from organic matter.
8. **Correct Answer (A) Using fuel-efficient vehicles**
   Using less fuel for the same distance reduces the overall consumption of non-renewable fossil fuels.
9. **Correct Answer (C) Sun**
   Coal is formed from ancient plants that captured energy from the Sun through photosynthesis millions of years ago.
10. **Correct Answer (D) Coal**
    Coal is a fossil fuel created from buried vegetation over geological timescales (millions of years).

WORKSHEET-2

1. **Correct Answer (B) Unwanted substances are added to air**
   Air pollution is defined as the presence of substances in the air that are harmful to humans, other living beings, or the environment.
2. **Correct Answer (C) Carbon monoxide**
   Carbon monoxide is a poisonous, flammable gas and a primary gaseous pollutant.
3. **Correct Answer (B) Breathing**
   Fine suspended particulate matter (SPM) can penetrate deep into the lungs, causing respiratory issues such as asthma and chronic obstructive pulmonary disease.
4. **Correct Answer (A) Carbon dioxide**
   Carbon dioxide is a major greenhouse gas that traps heat in the atmosphere.
5. **Correct Answer (B) Greenhouse gas emissions**
   The Kyoto Protocol is an international treaty aimed at reducing the emission of gases that contribute to global warming.

6. **Correct Answer (A) Marble and limestone Acid rain reacts with the calcium carbonate in marble and limestone, leading to a phenomenon often called "stone cancer" (as seen with the Taj Mahal).**
7. **Correct Answer (C) Nitrogen**
Nitrogen makes up approximately 78% of the Earth's atmosphere.
8. **Correct Answer (A) Vehicle exhaust**
is primarily released through the combustion of fossil fuels in motor vehicles and industrial plants.
9. **Correct Answer (B) Air pollution**
When sprayed, pesticides can drift into the air as vapours or small particles, contributing to atmospheric contamination.
10. **Correct Answer (C) Rotten eggs**
Hydrogen sulphide is famously characterized by its distinct, unpleasant odour of decaying organic matter or rotten eggs.

WORKSHEET-3

1. **Correct Answer (D) Physical, chemical and biological**

Water pollutants affect the appearance/temperature (physical), pH/composition (chemical), and the presence of microorganisms (biological) in water.

2. **Correct Answer (A) Textile and chemical industries**

These industries use heavy metals like lead and mercury in dyes and manufacturing processes, often discharging them as waste.

3. **Correct Answer (B) Cholera**

Cholera is a classic water-borne disease caused by ingesting water contaminated with the bacterium Vibrio cholerae.

4. **Correct Answer (C) Biomagnification**

This refers to the process where the concentration of toxins (like mercury) increases as you move up the food chain.

5. **Correct Answer (B) Prevent water pollution**
Tree roots act as natural filters that trap sediments and absorb excess nutrients/pollutants before they reach the river.
6. **Correct Answer (C) Water pollution**
By definition, the contamination of water bodies with harmful substances like sewage and industrial waste is water pollution.
7. **Correct Answer (C) Lead and mercury**
These specific industries are known for discharging toxic heavy metals like lead and mercury into water systems.
8. **Correct Answer (A) Microbes and bacteria**
Sewage is rich in organic waste, which provides a breeding ground for disease-causing bacteria and microbes.
9. **Correct Answer (B) Cholera**
As noted in question 3, cholera is directly linked to drinking or using polluted water.
10. **Correct Answer (B) Eutrophication**

This is the process where excess nutrients (like nitrates) lead to dense plant growth (algal blooms) and the death of animal life from lack of oxygen.

COMPETITIVE CORNER

SINGLE OPTION CORRECT TYPE

1. **Correct Answer (B) It combined waste management, sewage treatment, and the development of river fronts.**

The Ganga Action Plan was not just about one issue; it aimed to improve water quality through diverse strategies like intercepting sewage, treating industrial waste, and improving the environment along the banks.

2. **Correct Answer (D) It does not support respiration directly**

While nitrogen makes up 78% of the air we breathe, our bodies are not designed to process it for energy. We simply breathe it in and out unchanged.

3. **Correct Answer (C) They produce less pollution**
   CNG (Compressed Natural Gas) and LPG (Liquefied Petroleum Gas) are "cleaner" fuels because they burn more completely, leaving behind very little smoke or harmful residue.
4. **Correct Answer (B) They require millions of years to form**
   Fossil fuels are "non-renewable" because we are using them much faster than nature can replace them. The process of turning ancient plants into coal or oil takes millions of years under high pressure and heat.
5. **Correct Anwer (B) Decrease in fish population**
   Heavy algal growth (blooms) eventually dies and decomposes. This decomposition uses up the dissolved oxygen in the water, causing fish to suffocate.
6. **Correct Answer (C) They bypass nasal filtration and reach lungs**
   SPM (Suspended Particulate Matter) are so tiny that the hairs and mucus in our nose can't trap them. They travel deep into the lungs and even enter the bloodstream.
7. **Correct Answer (A) Carbon monoxide levels**
   Petrol engines release significant amounts of Carbon Monoxide (CO). CNG is a much cleaner-burning fuel, so switching to it immediately reduces the output of this toxic gas.
8. **Correct Answer (C) It decreases due to oxygen depletion**
   This describes the end stage of eutrophication. As oxygen disappears, most aquatic animals (fish, insects, etc.) cannot survive, leading to a loss of variety in life.
9. **Correct Answer (C) Rising sea levels**

   Greenhouse gases cause Global Warming, which melts polar ice caps and causes ocean water to expand, leading to rising sea levels that threaten coastal cities.
10. **Correct Answer (C) Fly ash**
    Particulate pollutants are solid or liquid particles suspended in the air. Fly ash consists of fine solid particles expelled from chimneys (usually from coal plants).

ONE OR MORE THAN ONE OPTION CORRECT TYPE

11. **Correct Answer A) Sun, B) Wind, D) Forest**

    Renewable resources are those that replenish naturally over a short period. The sun and wind are constant energy sources. Forests are also considered renewable because trees can be replanted and regrown.
12. **Correct Answer A) Smoke, B) Dust, D) Fly ash**

    Particulate pollutants (particulates) are microscopic solid or liquid particles suspended in the air. This includes smoke from fires, dust from construction or nature, and fly ash from coal-burning power plants.
13. **Correct Answer A) Treating industrial waste, B) Planting trees along river banks, D) Using biodegradable detergents**

    These are active prevention methods. Treatment removes toxins, trees act as natural filters for runoff, and biodegradable detergents break down naturally rather than lingering in water systems.
14. **Correct Answer A) They take millions of years to form, C) They cannot be replenished quickly**

    The definition of a non-renewable resource is one that is used faster than it can be replaced. Fossil fuels take millions of years to develop from ancient biomass, far exceeding the human timescale for replenishment.

15. **Correct Answer A) Rise in global temperature, B) Melting of polar glaciers, C) Flooding of coastal areas**

Increased greenhouse gases trap more heat, causing global warming. This leads to the melting of ice caps/glaciers, which in turn causes sea levels to rise and flood coastal regions.

ASSERTION AND REASONING TYPE

16. **Correct Answer D) A is false, but R is true**
17. **Correct Answer A) Both A and R are true, and R is the correct explanation of A**
18. **Correct Answer D) A is false, but R is true**
19. **Correct Answer A) Both A and R are true, and R is the correct explanation of A**

MATRIX MATCHING TYPE

20. **Correct Answer (A) a–Q, b–P, c–R, d–S.**
21. **Correct Answer (A) a–Q, b–R, c–S, d–P.**

COMPREHENSION TYPE

Passage-1

22. **Correct Answer (B) Reduces oxygen-carrying capacity of blood**
23. **Correct Answer (C) Human respiratory system**
24. **Correct Answer (C) Rainwater**

Passage-2

25. **Correct Answer (B) Excess nutrients**
26. **Correct Answer (C) Harmful microorganisms**
27. **Correct Answer (B) Reduction of dissolved oxygen**

PREVIOUS YEAR QUESTIONS

1. **Correct Answer (A) (i) Eutrophication, (ii) Cholera**

i) Eutrophication is the process where nutrient enrichment leads to algal blooms and oxygen depletion.
(ii) Cholera is a well-known water-borne disease.

2. **Correct Answer (D) Eutrophication- Phenomenon through which certain non-biodegradable pollutants get accumulated in tissue of organisms**

This is the mismatched pair because the definition provided
describes Biomagnification.
Eutrophication actually refers to nutrient enrichment leading to algal growth.

3. **Correct Answer (A) I and III only**

I (Bicycles) reduces vehicle emissions.

III (Unleaded petrol/CNG) produces significantly fewer pollutants than traditional fossil fuels.